

One-Way Sensitivity Analysis Report

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This one-way sensitivity analysis (OWSA) evaluates the robustness of the cost-effectiveness results for three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) compared to standard of care for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients for immunotherapy treatment.

The analysis examines how variations in individual model parameters affect the net monetary benefit (NMB) and optimal strategy selection, providing insights into which parameters have the greatest impact on decision uncertainty.

This report presents OWSA results for all three survival model structures:

- **Model A (Joint):** All biomarkers and treatment interactions in a single model
- **Model B (Focused):** Per-strategy formulas; CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction (adjusts for other biomarkers)
- **Model C (Separate):** Only one biomarker and its treatment interaction per model

This report contains:

- **Technical Details:** Analysis specifications and methodology
- **Parameter Overview:** Description of the key parameters varied in the sensitivity analysis
- **Model A Results:** Tornado diagrams, line plots, and impact rankings for the Joint model
- **Model B Results:** Tornado diagrams, line plots, and impact rankings for the Focused model
- **Model C Results:** Tornado diagrams, line plots, and impact rankings for the Separate model
- **Cross-Model Comparison:** Assessment of robustness across model structures
- **Survival Model Structural Sensitivity:** Impact of parametric distribution choice

2 Technical Details

2.1 Analysis Specifications

Model Structures Analyzed:

	Model	Name	Formula Pattern	Status
A	A	Joint (All-in-One)	Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx	Results available
B	B	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx	Results available
C	C	Separate	Age + sex + Rx + [biomarker]:Rx	Results available

Model Configuration (per model structure):

- Strategies evaluated: 4 (control, crp, tlr, tmb_braf)
- Parameters tested: 14
- Time horizon: 520 weeks (10 years)
- Discount rate: 4% per annum
- Willingness-to-pay threshold: €51,000
- Parameter variation range: $\pm 20\%$

Parameter Constraints:

- Utility parameters capped at maximum value of 1.0
- All cost parameters constrained to non-negative values

Methodology:

- One-way sensitivity analysis varies each parameter individually
- All other parameters held at base case values during each test
- DSA is run separately for each model structure to capture structural differences
- Net Monetary Benefit calculated as: $NMB = \text{Effect} \times WTP - \text{Cost}$
- Tornado diagrams ordered by magnitude of parameter impact
- Line plots show outcome response across actual parameter values
- Three outcome measures: Net Monetary Benefit, Total Costs, and Total Effectiveness

2.2 Computational Environment

Analysis completed on: mars 03, 2026 **R version:** R version 4.5.1 (2025-06-13 ucrt) **Repository:** ben-geisler/METIMMOX-1 **Report version:** 3.1

3 Analysis Parameters

Table 1: Parameters Included in One-Way Sensitivity Analysis

Group	Parameter Description	Base	Min (-20%)	Max (+20%)
Drug costs	Nivolumab cost per administration	13923	11138	16708
	FLOX chemotherapy cost per administration	427	342	512
Test costs	Next-generation sequencing (TMB/BRAF testing)	2518	2014	3022
	Computed tomography imaging	386	309	463
	Blood test cost	16.00	12.80	19.20
Other costs	Standard outpatient visit	33.00	26.40	39.60
	Comprehensive baseline visit	530	424	636
	Follow-up visit cost	33.00	26.40	39.60
	End-of-life care costs	13803	11042	16564
Utilities	Utility - progression-free state	0.73	0.58	0.88
	Utility - progressed disease state	0.59	0.47	0.71
Prevalence	CRP biomarker positive prevalence	0.34	0.27	0.41
	TLR biomarker positive prevalence	0.65	0.52	0.78
	TMB/BRAF biomarker positive prevalence	0.45	0.36	0.54
Survival model	Parametric survival distribution (OS & PFS)	gamma	See section 7	See section 7

4 Model A: Joint Model Results

Model A uses a **joint survival model** with all three biomarkers and all treatment-biomarker interactions included in a single model.

Formula: `Surv(time, event) ~ Age + sex + Rx + crp*Rx + tlr*Rx + tmb_braf*Rx`

4.1 Tornado Diagrams - Model A

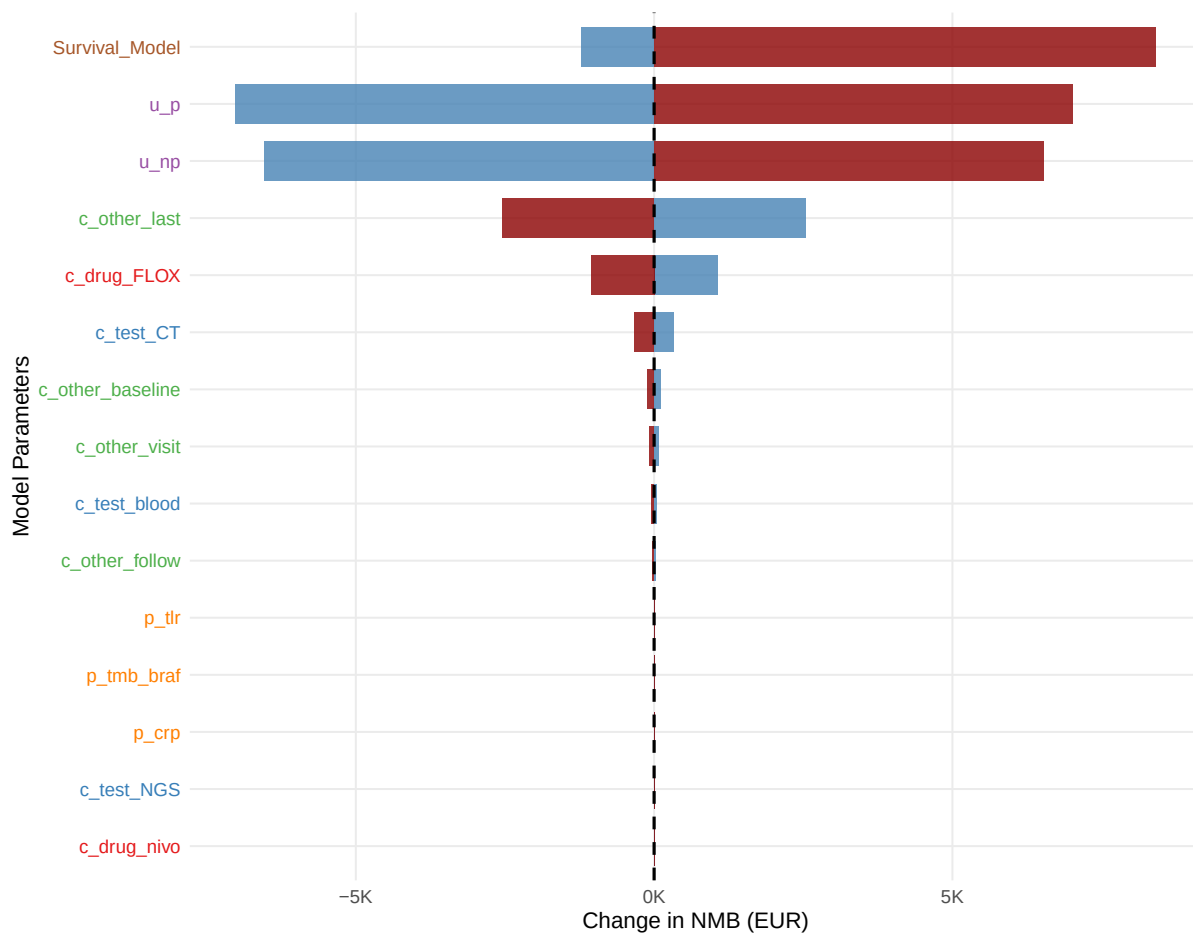


Figure 1: Tornado Diagram - Model A: Standard of Care

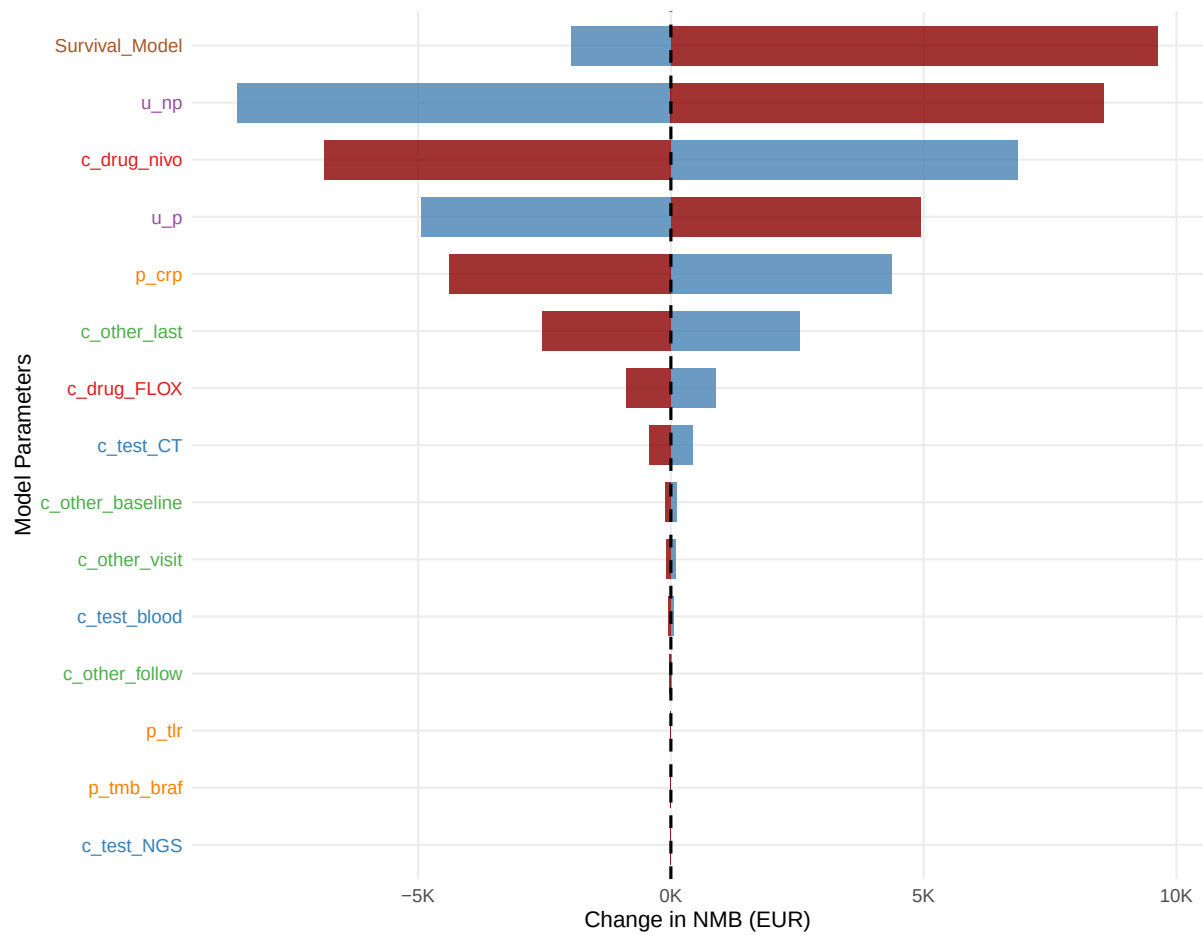


Figure 2: Tornado Diagram - Model A: CRP-guided Strategy

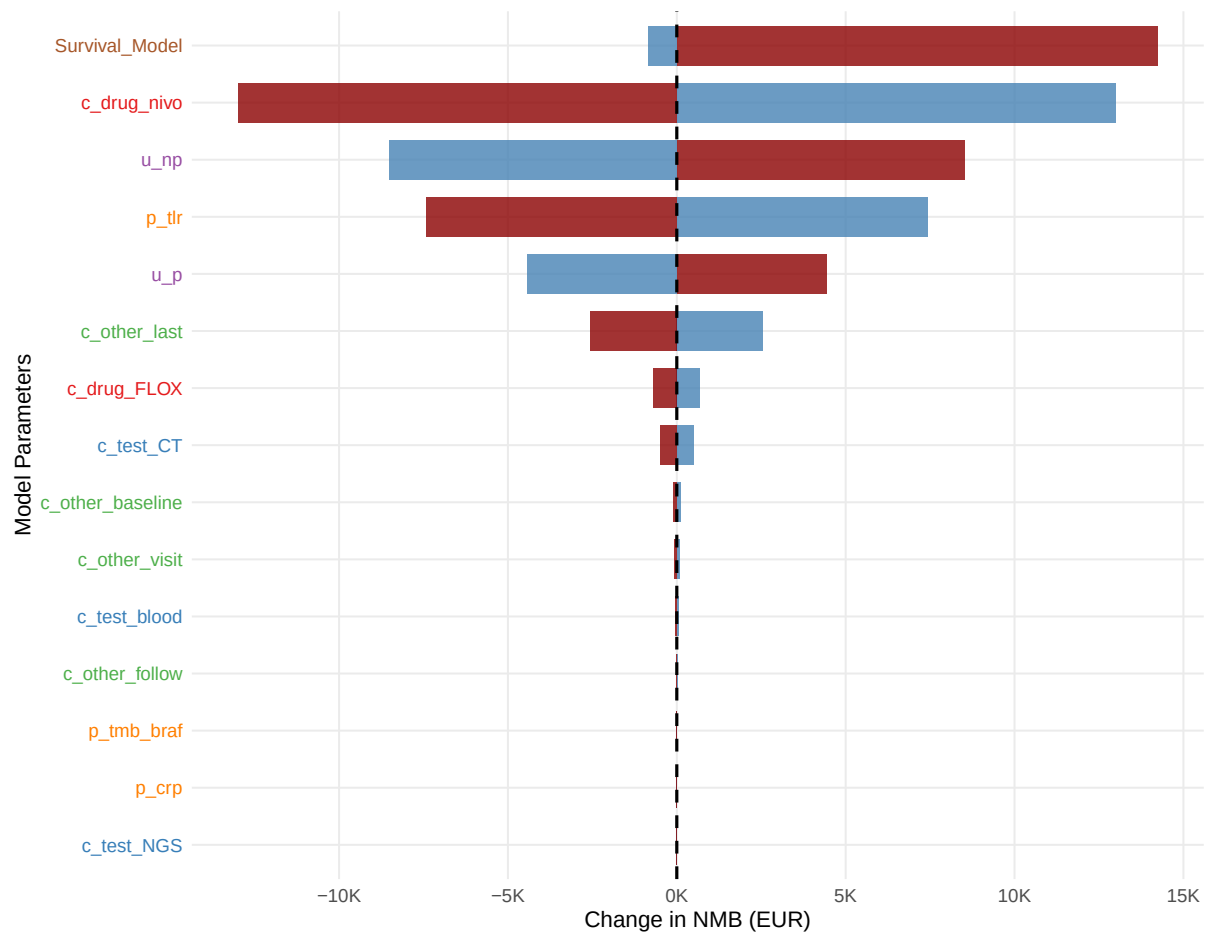


Figure 3: Tornado Diagram - Model A: TLR-guided Strategy

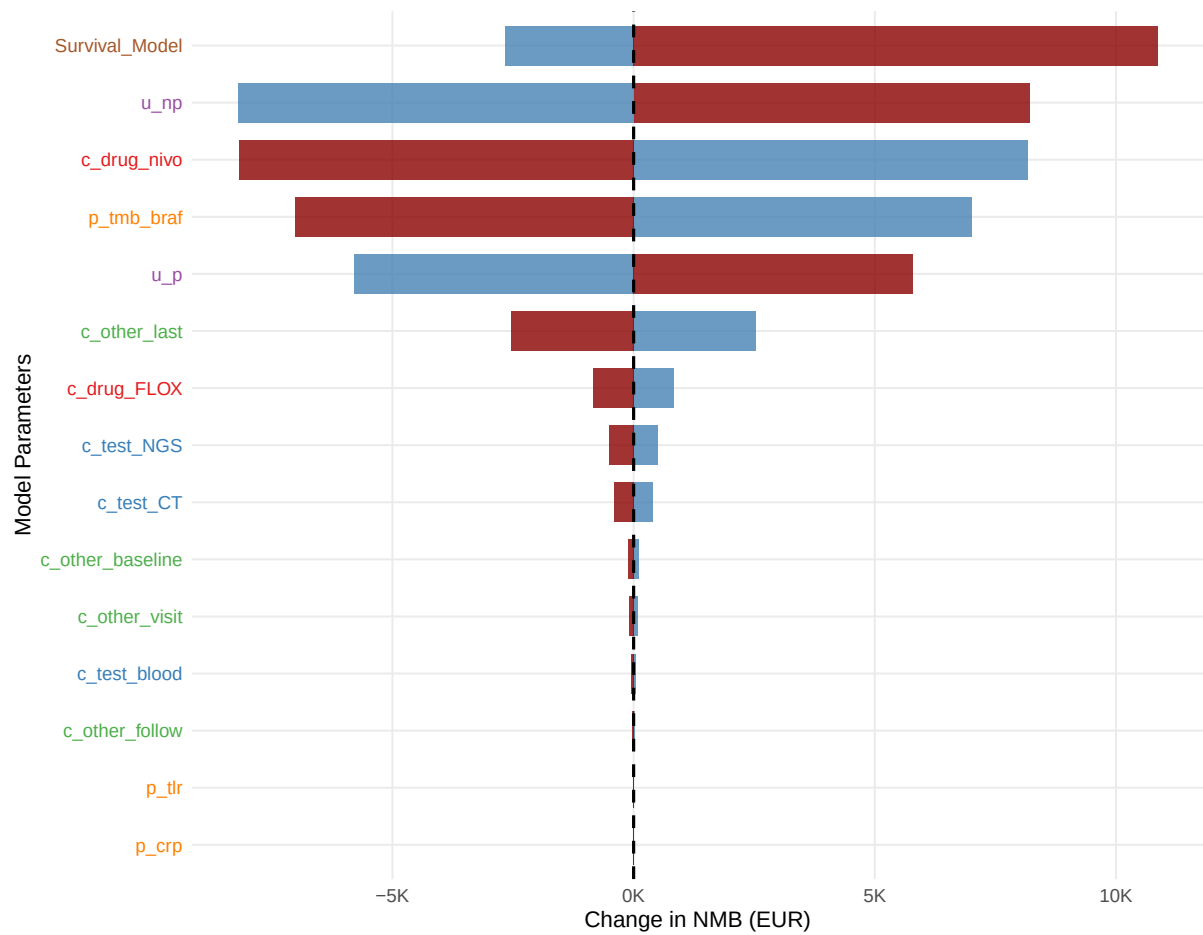


Figure 4: Tornado Diagram - Model A: TMB/BRAF-guided Strategy

4.2 Line Plot Sensitivity - Model A

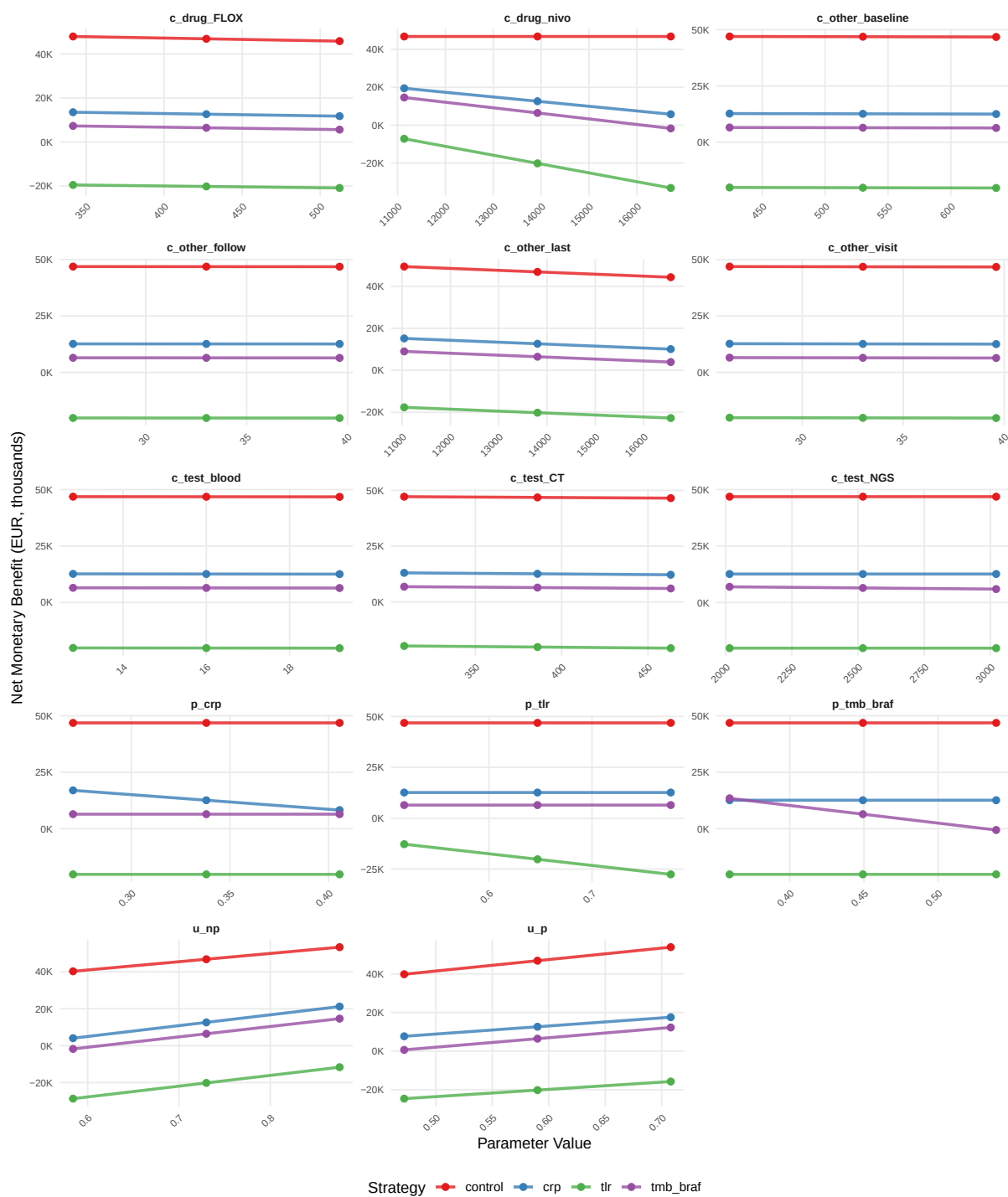


Figure 5: Model A: Parameter Sensitivity - Net Monetary Benefit

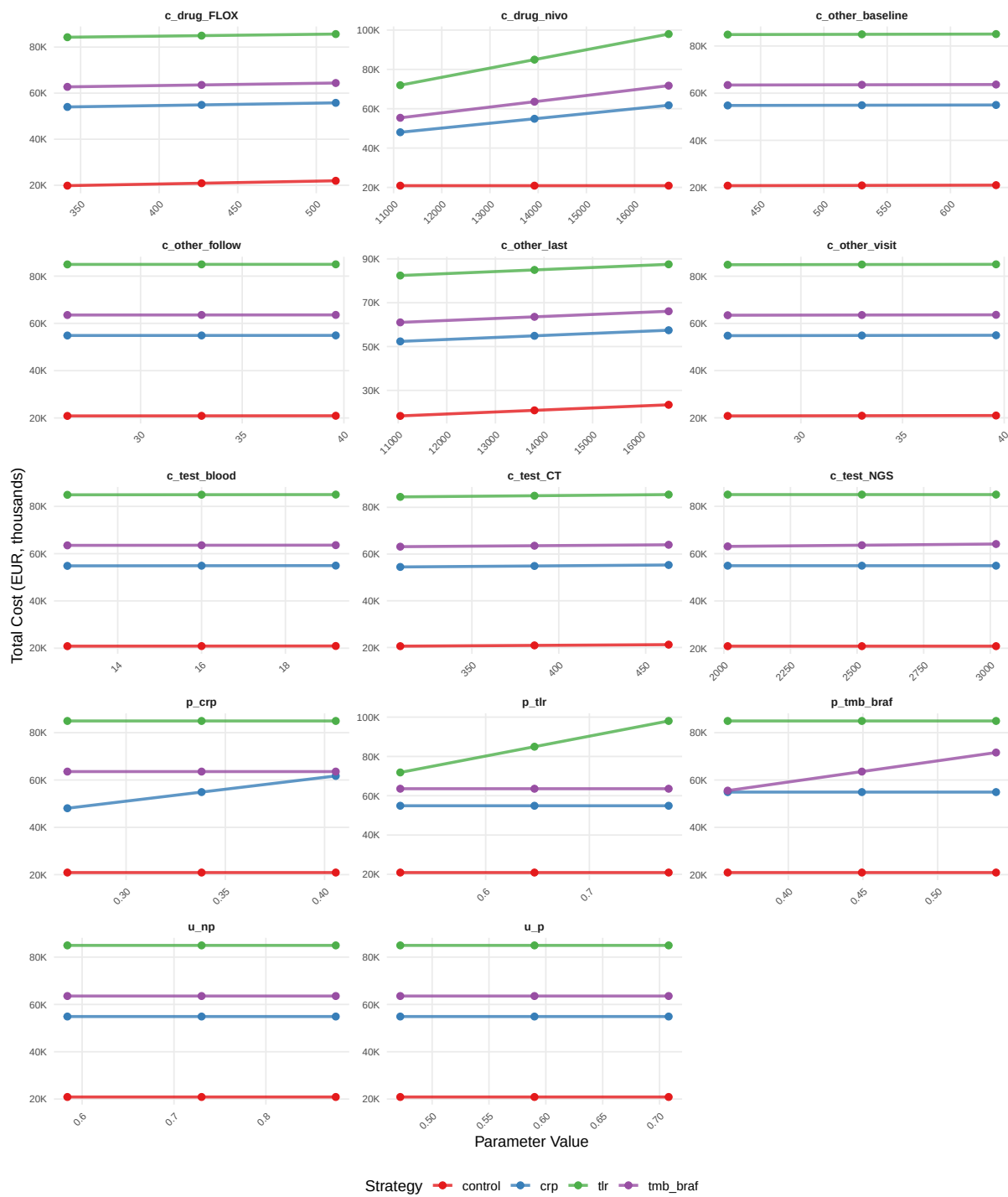


Figure 6: Model A: Parameter Sensitivity - Total Costs

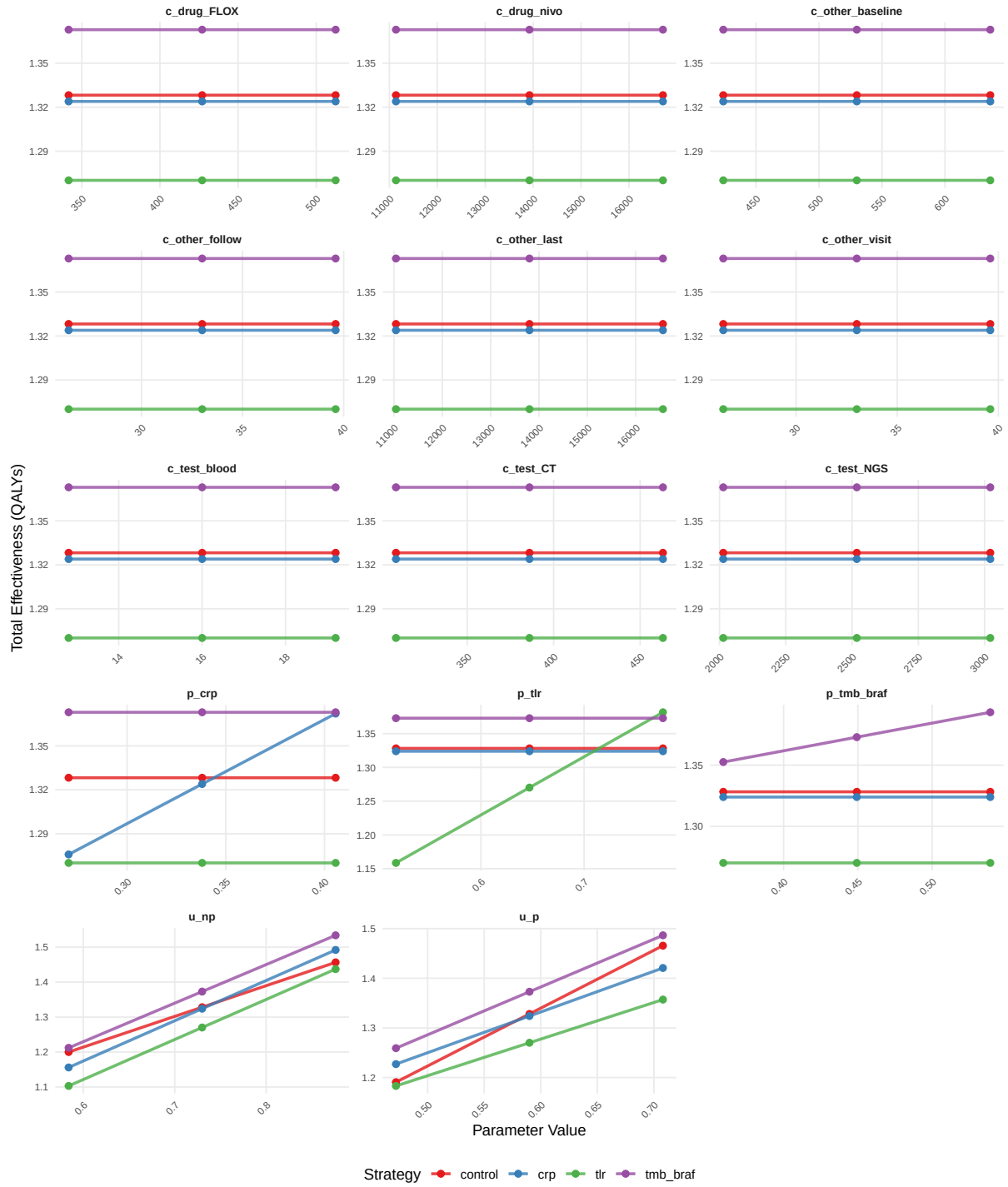


Figure 7: Model A: Parameter Sensitivity - Total Effectiveness

4.3 Parameter Impact Rankings - Model A

Table 3: Model A: Top 3 Most Influential Parameters by Strategy

Strategy	Rank	Parameter	NMB Range (EUR K)
	1	NA	9.6

CRP-guided	2	Utility (PF)	8.6
	3	Nivolumab cost	6.9
Standard of Care	1	NA	8.4
	2	Utility (P)	7.0
	3	Utility (PF)	6.5
TLR-guided	1	NA	14.2
	2	Nivolumab cost	13.0
	3	Utility (PF)	8.5
TMB/BRAF-guided	1	NA	10.9
	2	Utility (PF)	8.2
	3	Nivolumab cost	8.2

5 Model B: Focused Model Results

Model B uses **focused survival models** with per-strategy formulas. CRP and TMB/BRAF strategies use both interaction terms ($\text{crp:Rx} + \text{tmb_braf:Rx}$), while TLR uses only the TLR-treatment interaction (tlr:Rx), adjusting for other biomarkers as main effects.

Formulas:

- CRP/TMB_BRAF: $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{crp:Rx} + \text{tmb_braf:Rx}$
- TLR: $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{tlr:Rx}$

5.1 Tornado Diagrams - Model B

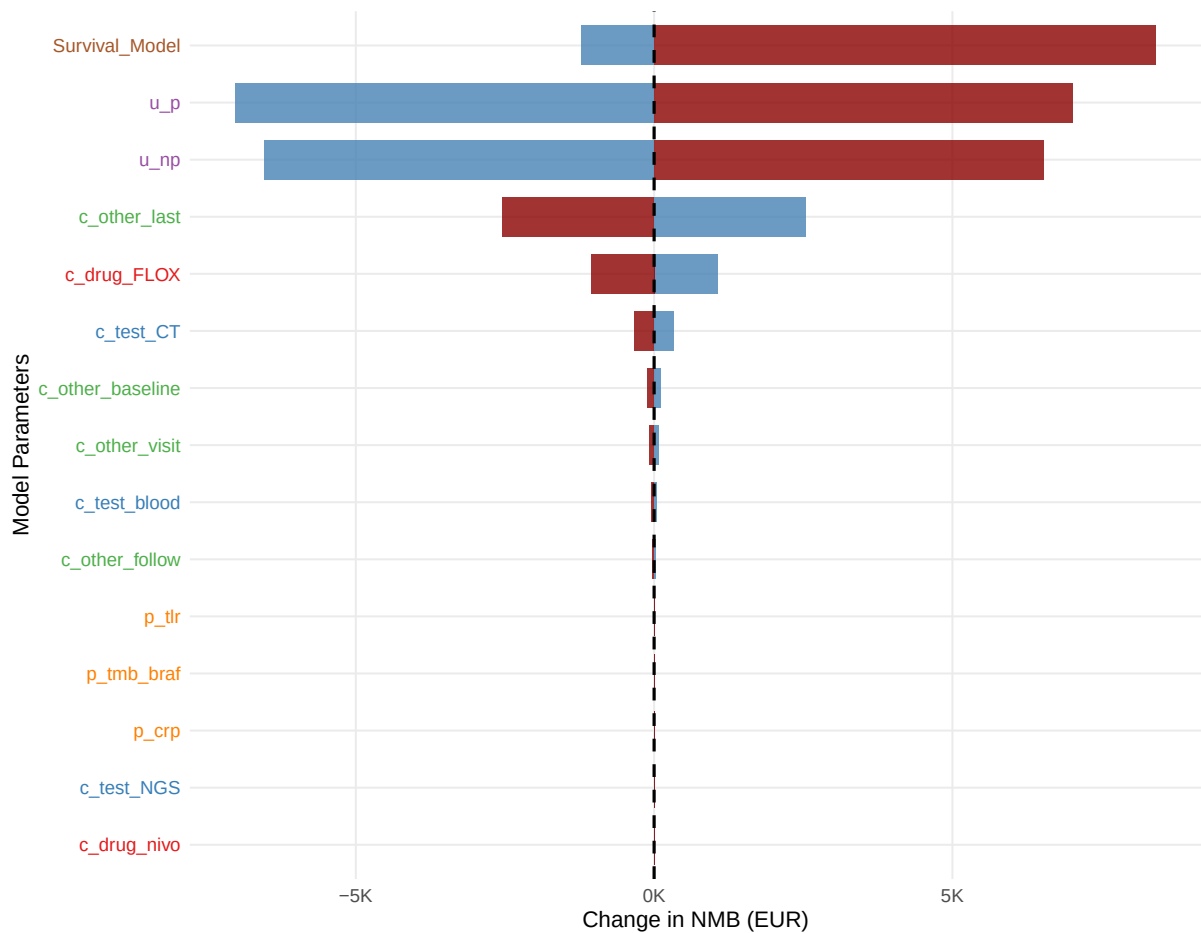


Figure 8: Tornado Diagram - Model B: Standard of Care

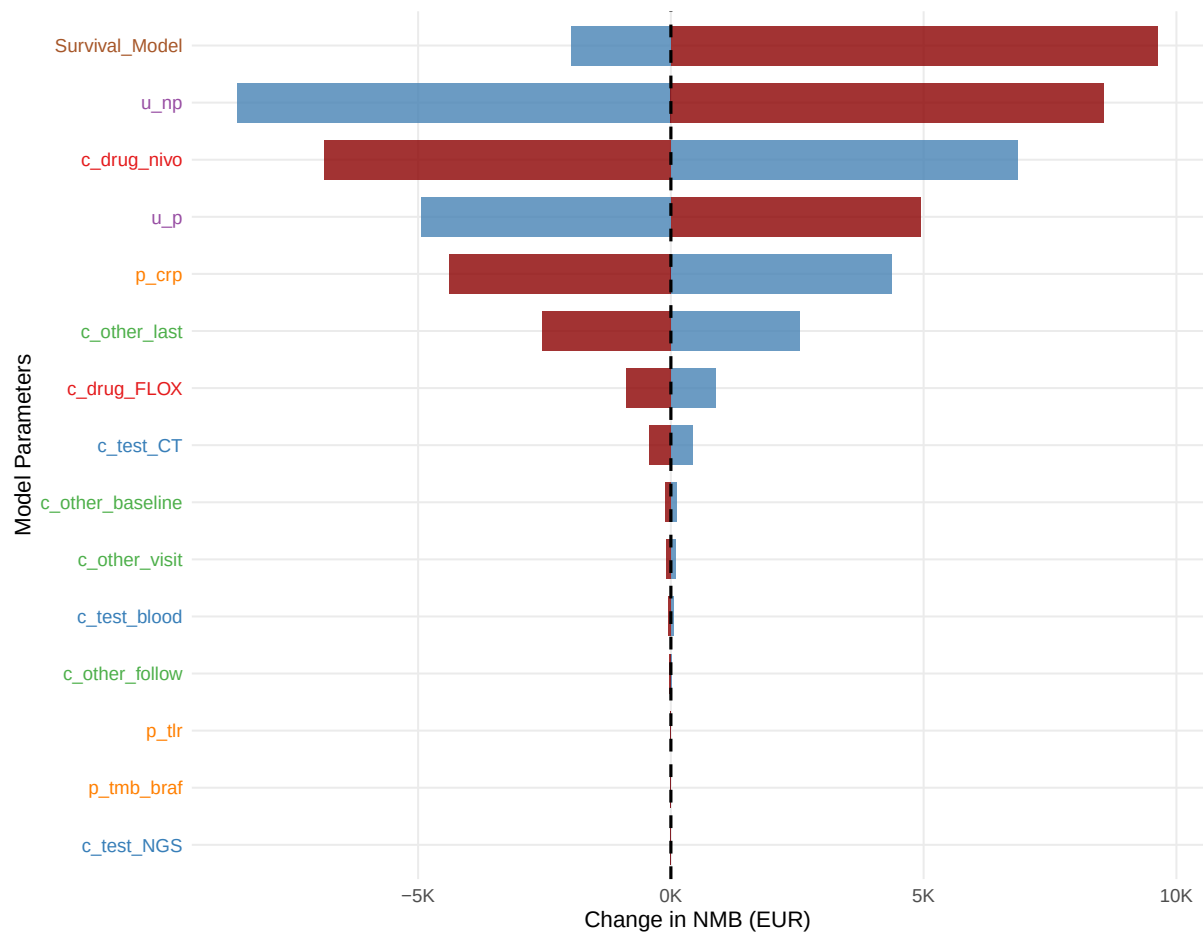


Figure 9: Tornado Diagram - Model B: CRP-guided Strategy

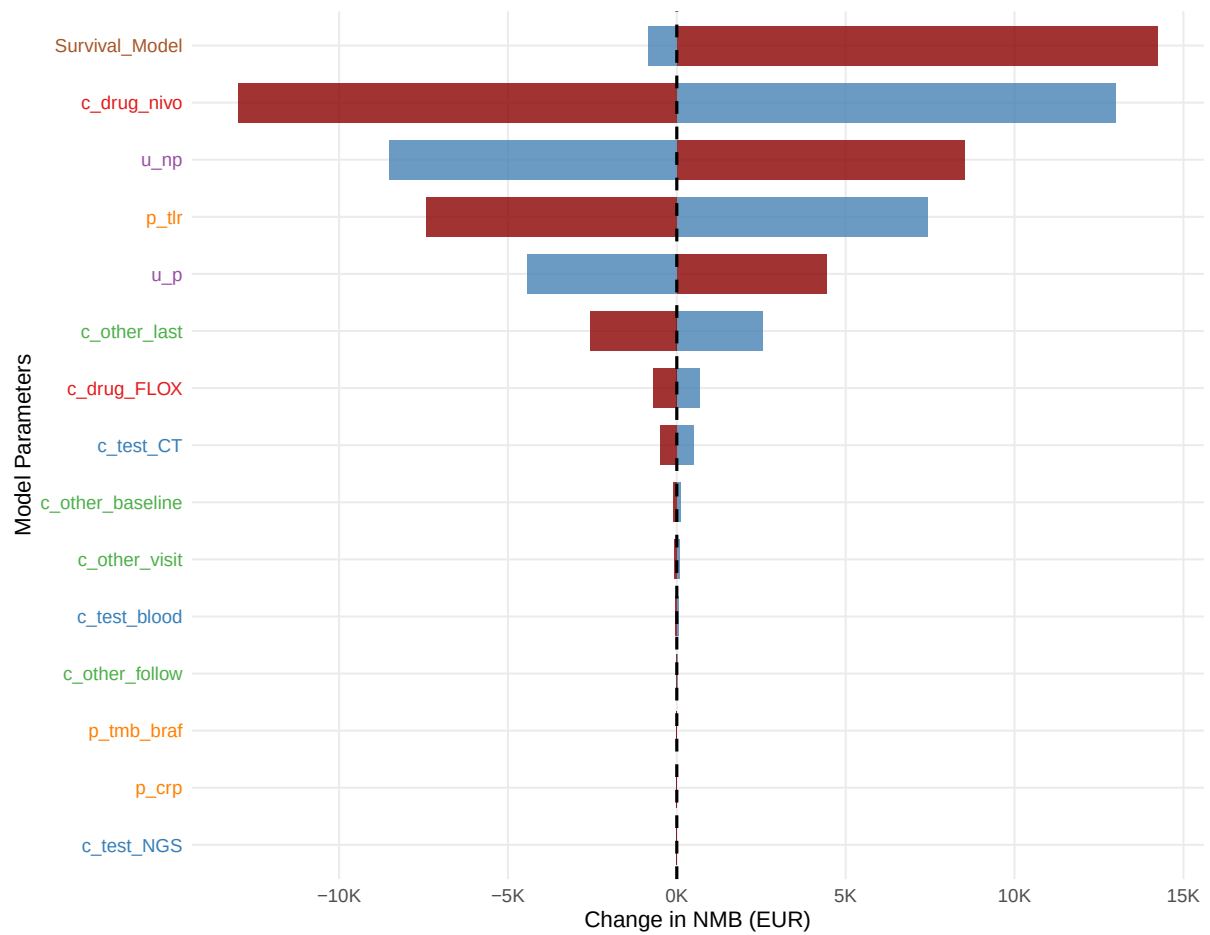


Figure 10: Tornado Diagram - Model B: TLR-guided Strategy

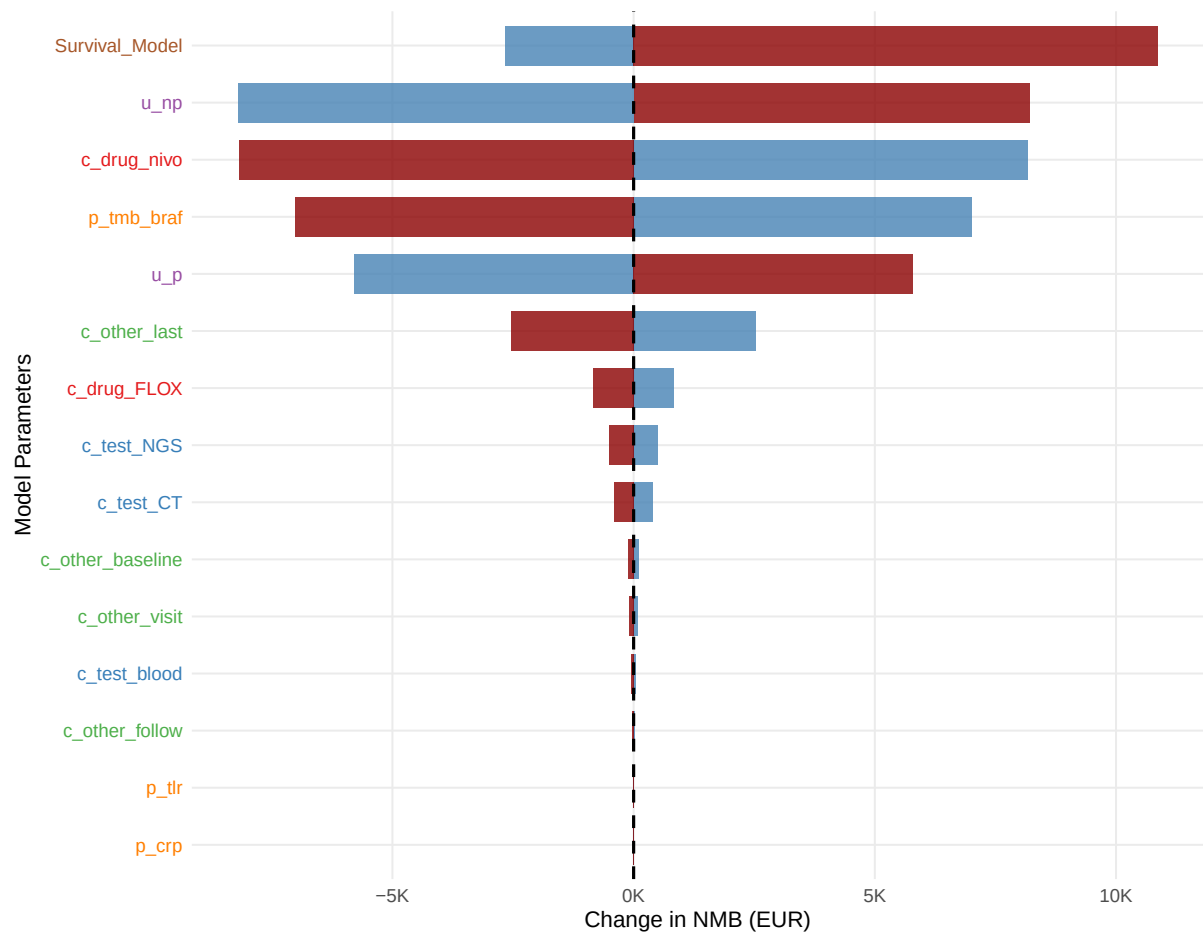


Figure 11: Tornado Diagram - Model B: TMB/BRAF-guided Strategy

5.2 Line Plot Sensitivity - Model B

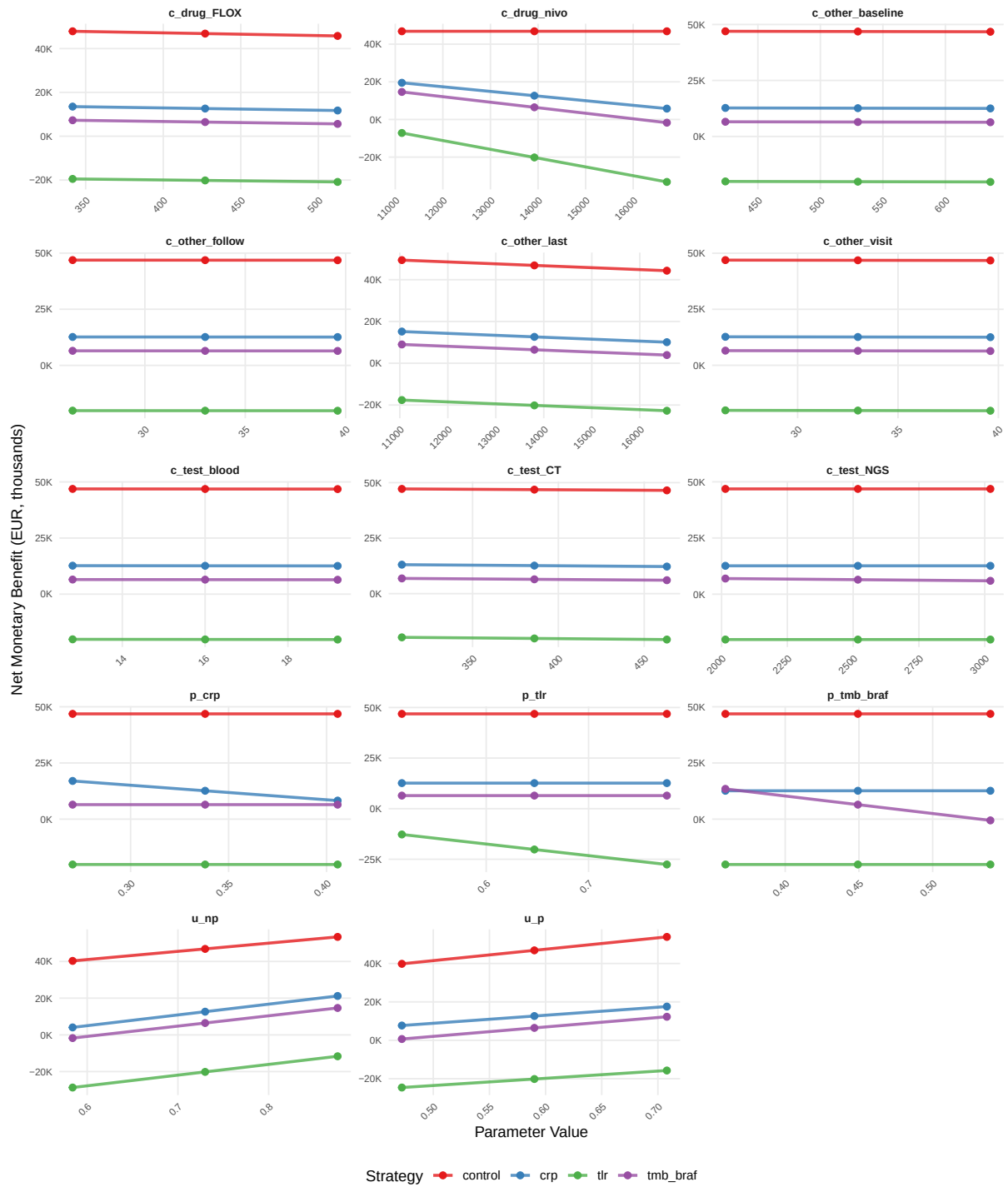


Figure 12: Model B: Parameter Sensitivity - Net Monetary Benefit

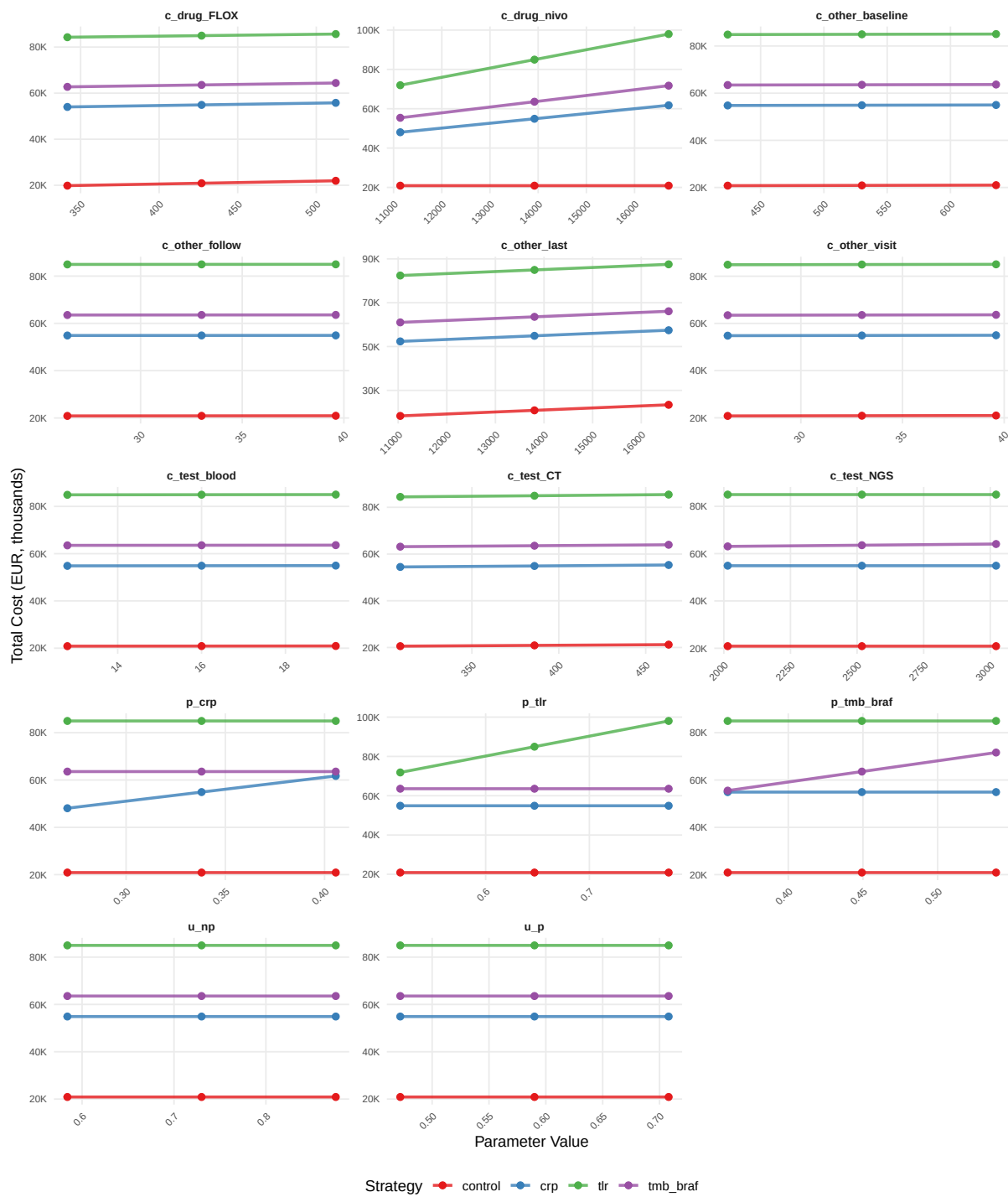


Figure 13: Model B: Parameter Sensitivity - Total Costs

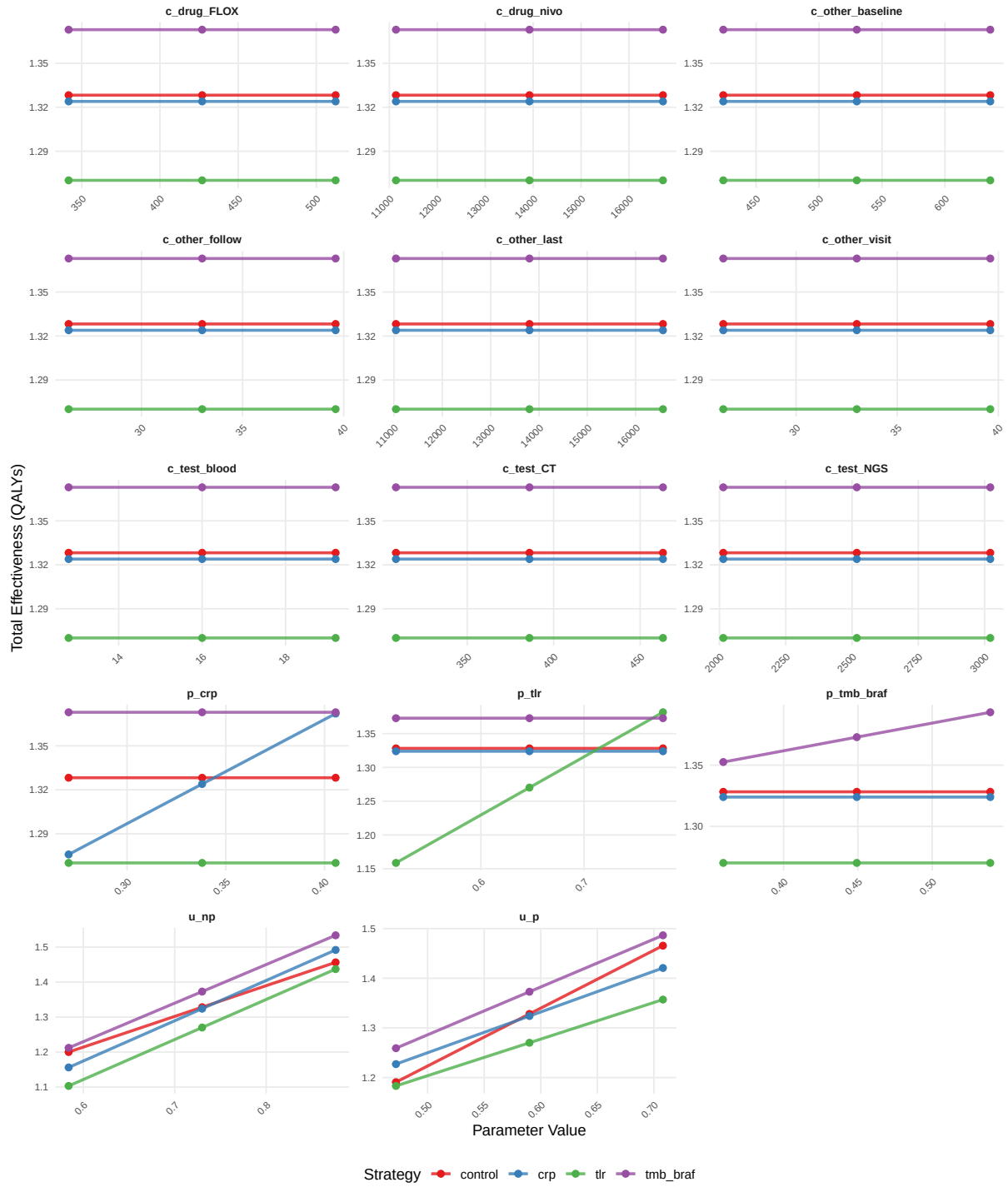


Figure 14: Model B: Parameter Sensitivity - Total Effectiveness

5.3 Parameter Impact Rankings - Model B

Table 4: Model B: Top 3 Most Influential Parameters by Strategy

Strategy	Rank	Parameter	NMB Range (EUR K)
	1	NA	9.6

CRP-guided	2	Utility (PF)	8.6
	3	Nivolumab cost	6.9
Standard of Care	1	NA	8.4
	2	Utility (P)	7.0
	3	Utility (PF)	6.5
TLR-guided	1	NA	14.2
	2	Nivolumab cost	13.0
	3	Utility (PF)	8.5
TMB/BRAF-guided	1	NA	10.9
	2	Utility (PF)	8.2
	3	Nivolumab cost	8.2

6 Model C: Separate Model Results

Model C uses **separate survival models** with only one biomarker and its treatment interaction per model.

Formula: `Surv(time, event) ~ Age + sex + Rx + [biomarker]:Rx`

6.1 Tornado Diagrams - Model C

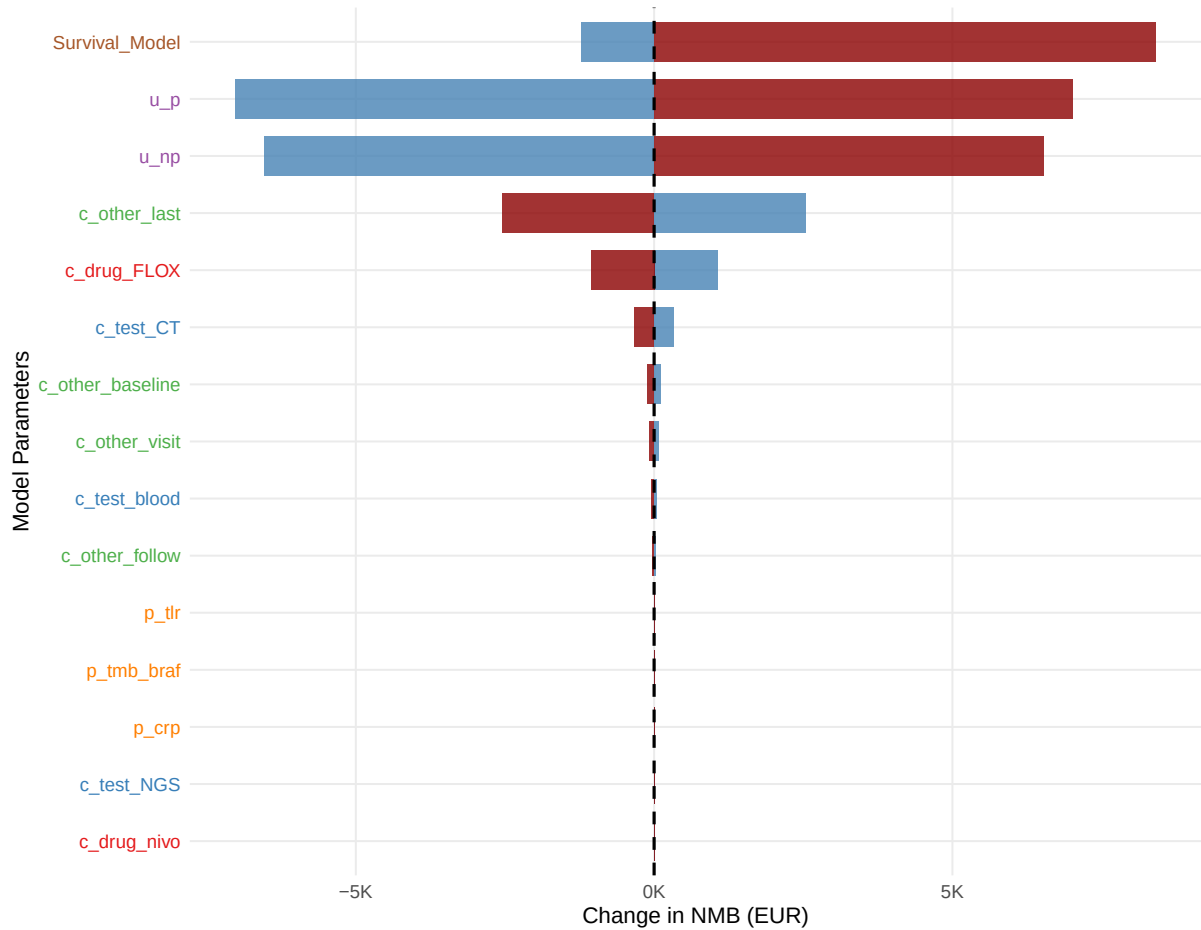


Figure 15: Tornado Diagram - Model C: Standard of Care

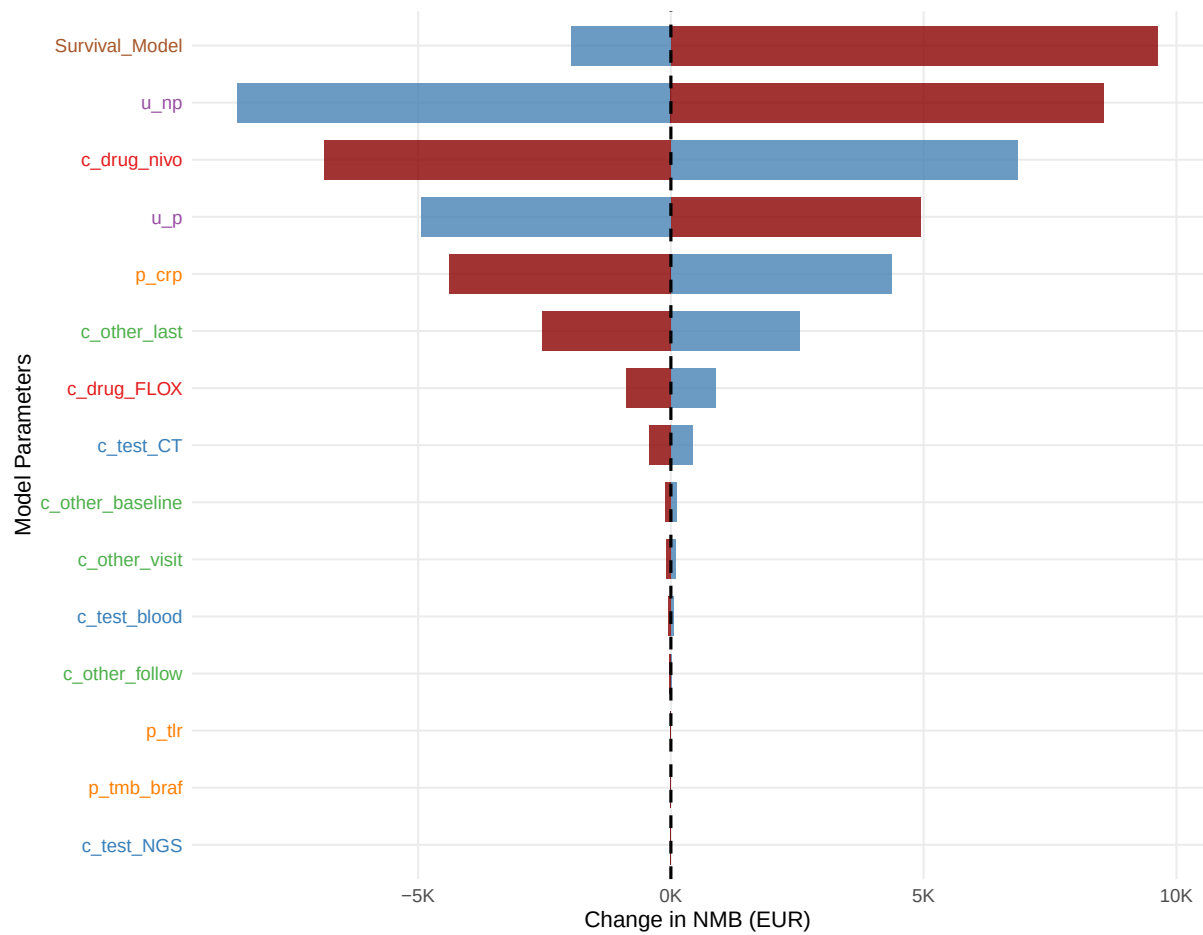


Figure 16: Tornado Diagram - Model C: CRP-guided Strategy

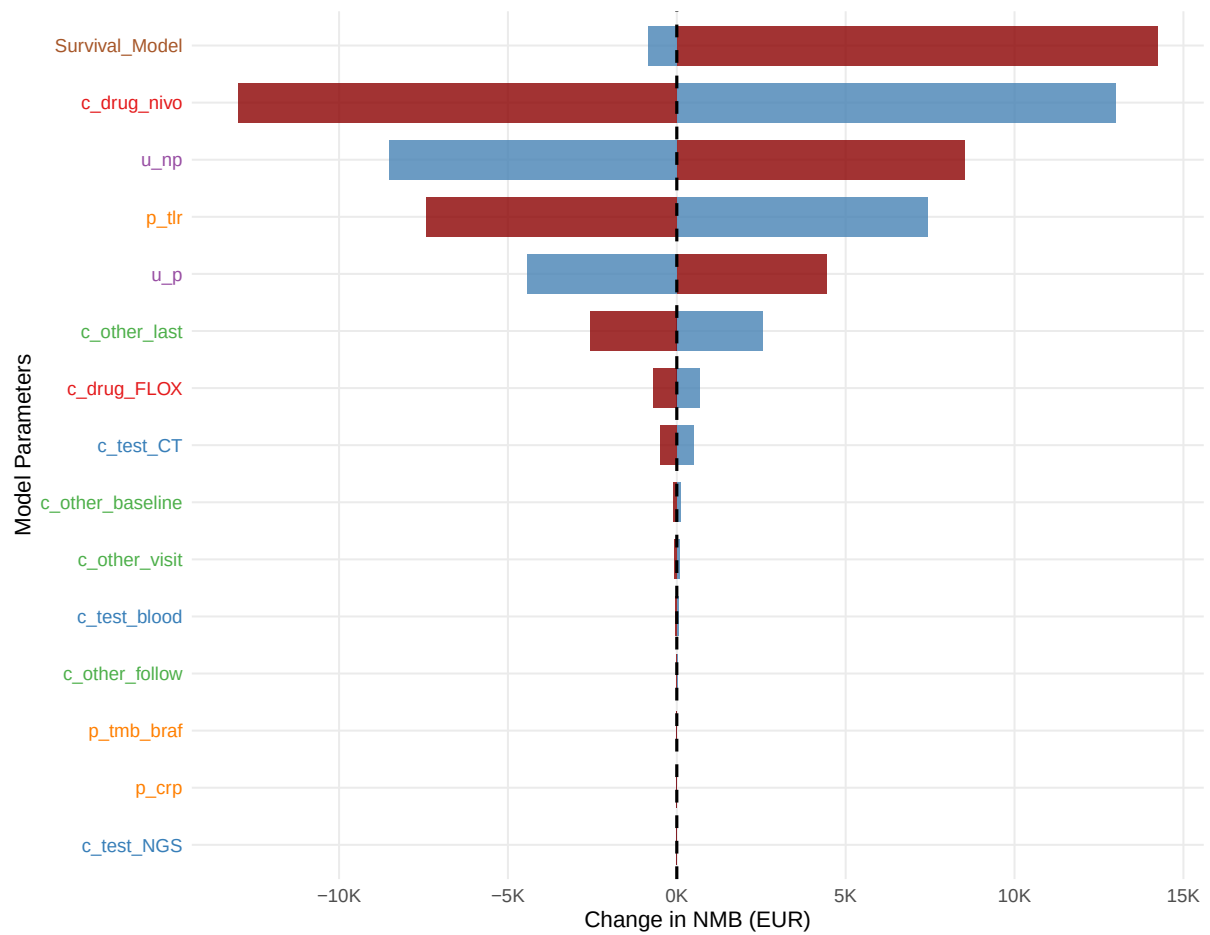


Figure 17: Tornado Diagram - Model C: TLR-guided Strategy

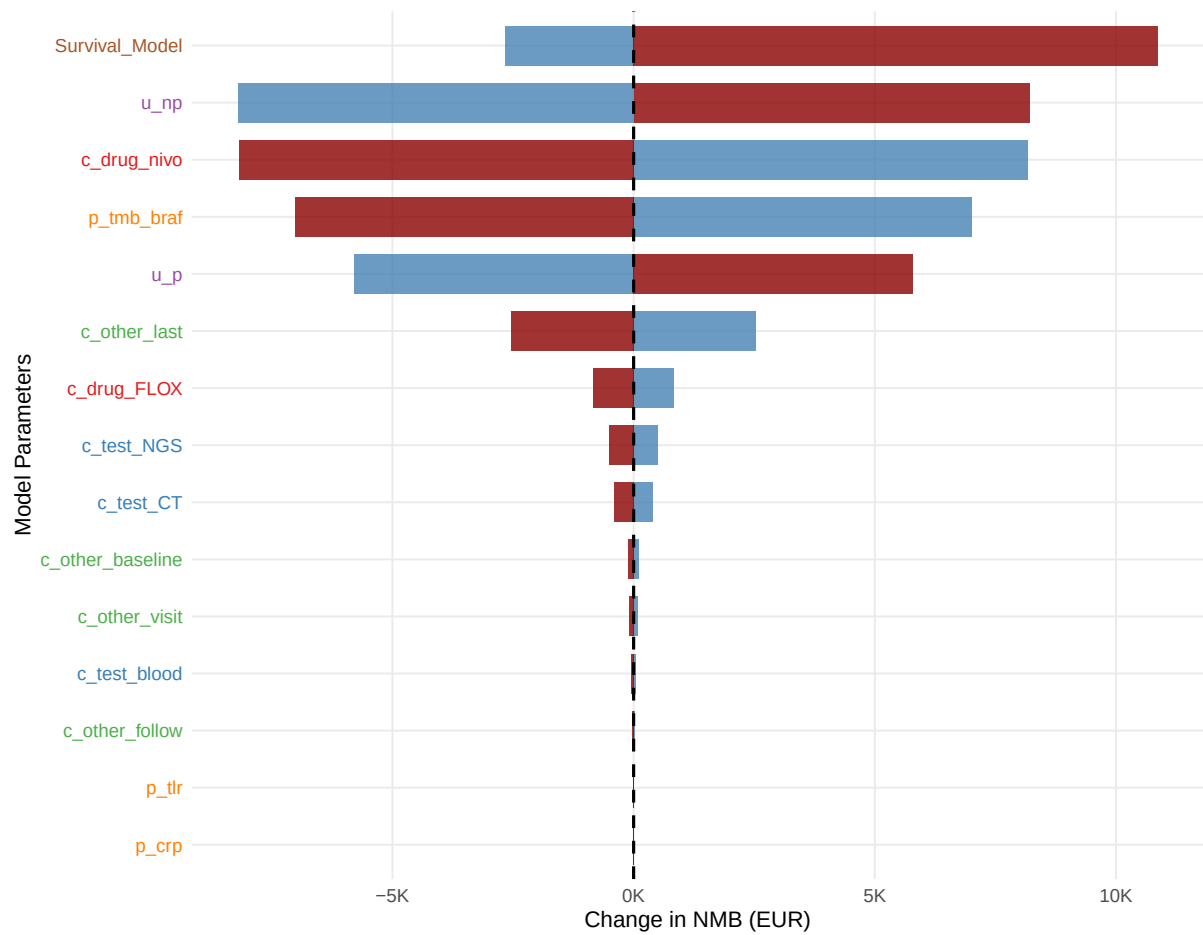


Figure 18: Tornado Diagram - Model C: TMB/BRAF-guided Strategy

6.2 Line Plot Sensitivity - Model C

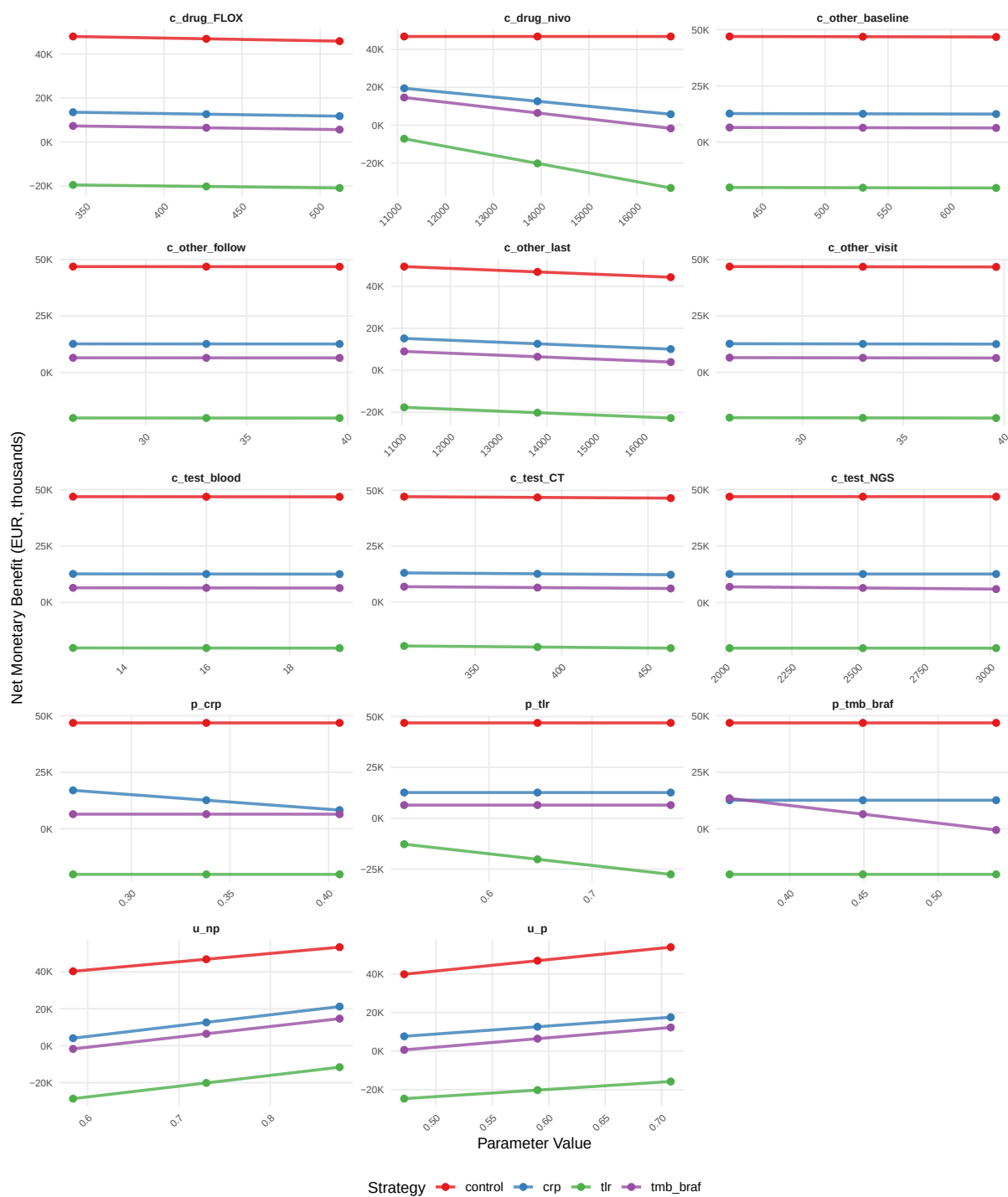


Figure 19: Model C: Parameter Sensitivity - Net Monetary Benefit

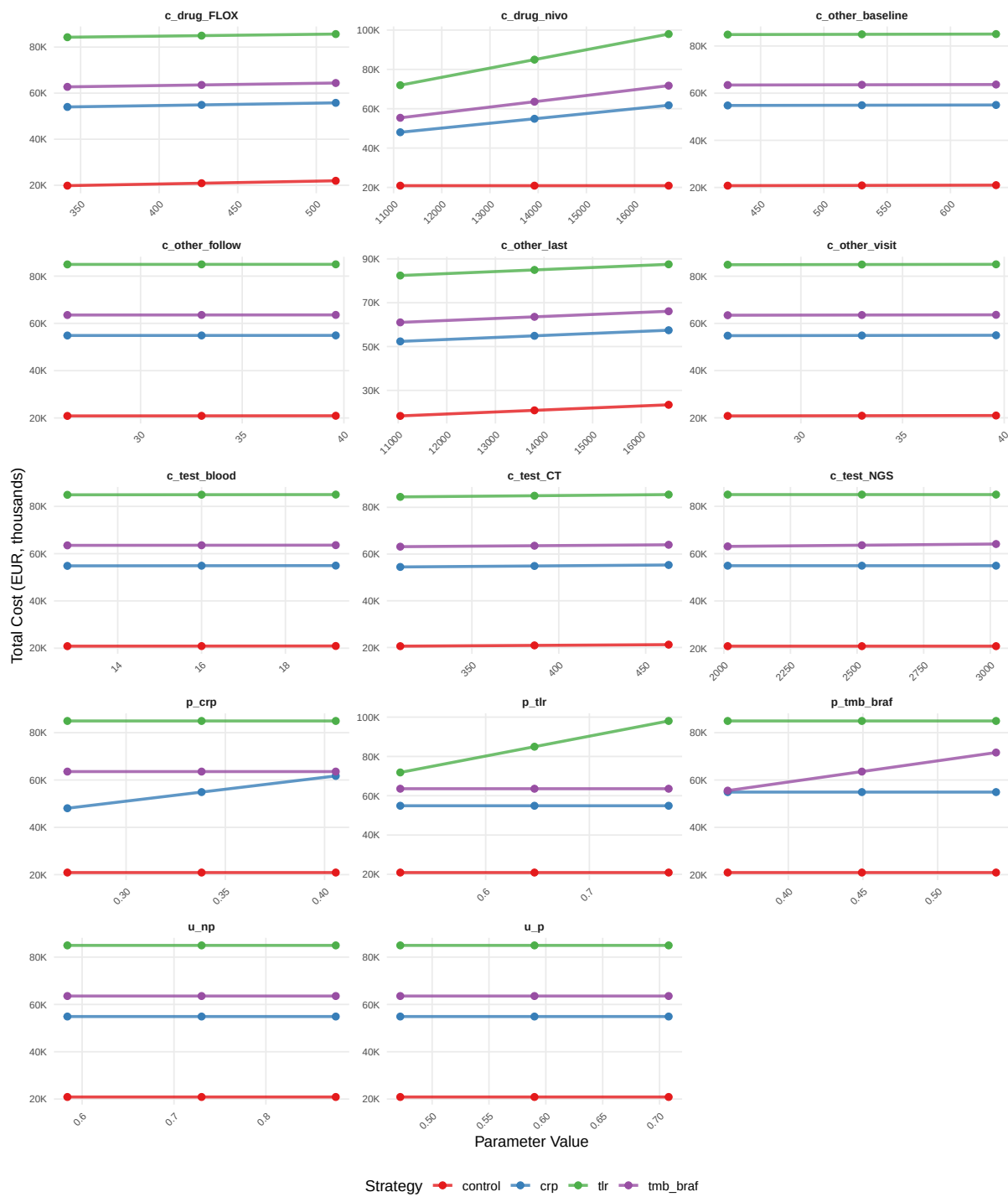


Figure 20: Model C: Parameter Sensitivity - Total Costs

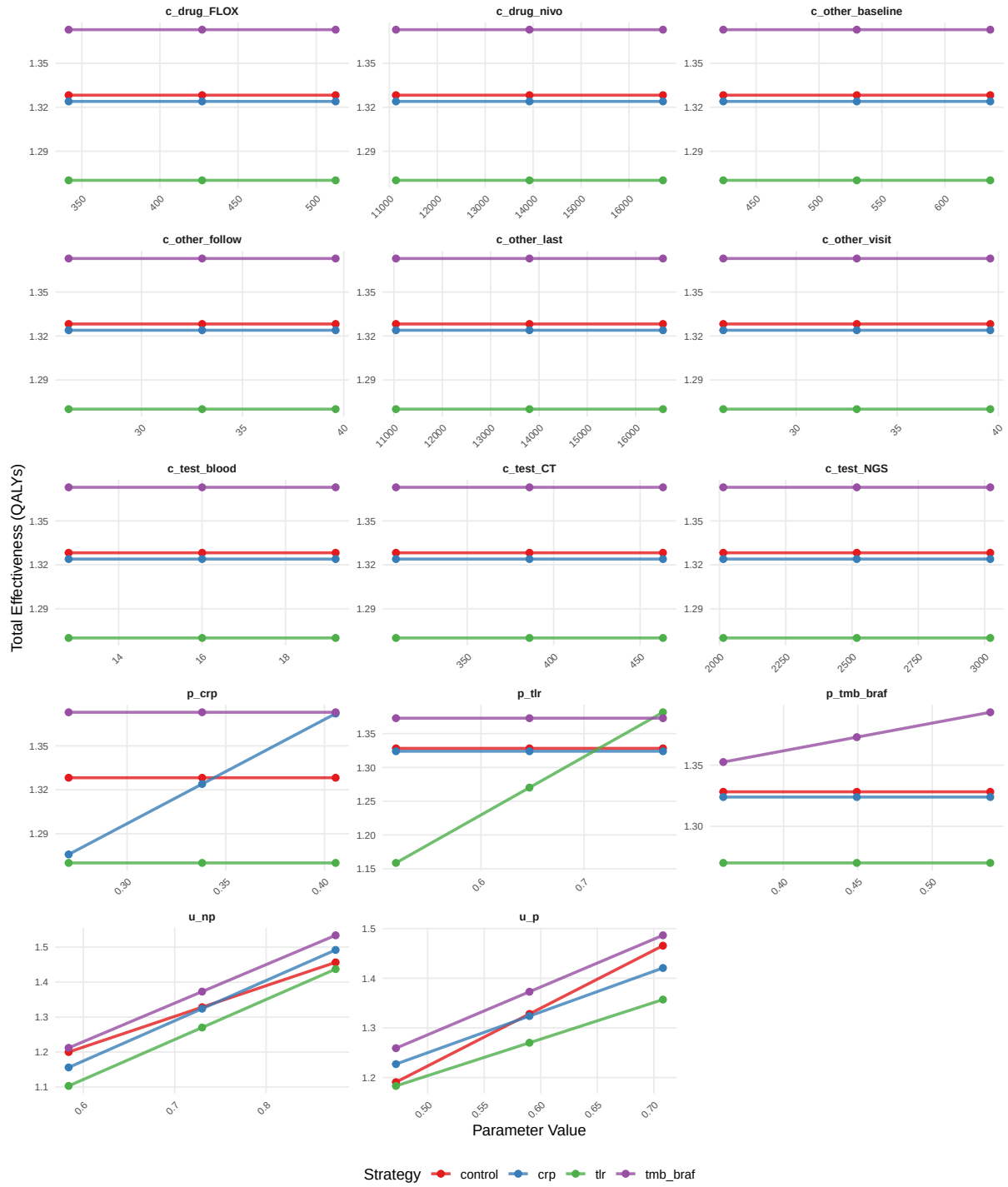


Figure 21: Model C: Parameter Sensitivity - Total Effectiveness

6.3 Parameter Impact Rankings - Model C

Table 5: Model C: Top 3 Most Influential Parameters by Strategy

Strategy	Rank	Parameter	NMB Range (EUR K)
	1	NA	9.6

CRP-guided	2	Utility (PF)	8.6
	3	Nivolumab cost	6.9
Standard of Care	1	NA	8.4
	2	Utility (P)	7.0
	3	Utility (PF)	6.5
TLR-guided	1	NA	14.2
	2	Nivolumab cost	13.0
	3	Utility (PF)	8.5
TMB/BRAF-guided	1	NA	10.9
	2	Utility (PF)	8.2
	3	Nivolumab cost	8.2

7 Cross-Model Comparison

This section compares the sensitivity analysis results across all three model structures to assess the robustness of conclusions.

7.1 Optimal Strategy Comparison

Table 6: Base Case Optimal Strategy by Model Structure

	Model	Structure	Optimal Strategy
A	A (Joint)	All interactions in one model	control
B	B (Focused)	All biomarkers + one interaction	control
C	C (Separate)	One biomarker per model	control

7.2 Top Parameter Comparison

Table 7: Most Influential Parameter by Strategy and Model (with NMB Range in EUR K)

Strategy	Model A	Model B	Model C
Standard of Care	Survival_Model (8.4K)	Survival_Model (8.4K)	Survival_Model (8.4K)
CRP-guided	Survival_Model (9.6K)	Survival_Model (9.6K)	Survival_Model (9.6K)
TLR-guided	Survival_Model (14.2K)	Survival_Model (14.2K)	Survival_Model (14.2K)
TMB/BRAF-guided	Survival_Model (10.9K)	Survival_Model (10.9K)	Survival_Model (10.9K)

7.3 Robustness Assessment

Robust Conclusion: The optimal strategy (**control**) is consistent across all 3 model structures analyzed.

This indicates that the cost-effectiveness conclusions are robust to the choice of survival model specification.

8 Survival Model Structural Sensitivity

8.1 Overview

This section examines how the choice of parametric survival distribution affects cost-effectiveness results. The base case uses the best-fitting distribution selected by AIC. All 9 fitted distributions are tested with the constraint that OS and PFS use the same distribution to ensure methodological consistency.

Distributions tested:

1. Exponential
2. Weibull
3. Weibull PH (proportional hazards)
4. Log-logistic
5. Log-normal
6. Gamma (base case)
7. Gompertz
8. Generalized Gamma
9. Generalized F

8.2 NMB by Parametric Distribution

Table 3: Net Monetary Benefit by Parametric Distribution and Strategy

Distribution	control	crp	tlr	tmb_braf	control Rank	crp Rank	tlr Rank	tmb_braf Rank
exponential	55257	22258	-5942	17321	1.0	1.0	1.0	1.0
weibull	49255	12071	-21015	5528	2.5	7.5	8.5	7.5
weibullph	49255	12071	-21015	5528	2.5	7.5	8.5	7.5
llogis	45658	15987	-17565	9285	8.0	2.0	3.0	2.0
lognormal	45631	14636	-16307	8327	9.0	3.0	2.0	3.0
gamma	46848	12627	-20182	6456	5.0	6.0	4.0	6.0
gompertz	49084	10656	-20624	3785	4.0	9.0	7.0	9.0
gengamma	46708	12652	-20383	6737	7.0	4.0	6.0	5.0
genf	46781	12637	-20295	6745	6.0	5.0	5.0	4.0

8.3 Impact Summary

Table 4: Survival Model Impact on NMB by Strategy

Strategy	Base NMB	Min NMB	Max NMB	Range
Standard of Care	46848	45631 (lognormal)	55257 (exponential)	9626
CRP-guided	12627	10656 (gompertz)	22258 (exponential)	11602
TLR-guided	-20182	-21015 (weibullph)	-5942 (exponential)	15073
TMB/BRAF-guided	6456	3785 (gompertz)	17321 (exponential)	13536

8.4 Interpretation

The survival model choice represents **structural uncertainty** in the cost-effectiveness analysis. Different parametric distributions make different assumptions about how hazard rates change over time:

- **Exponential:** Constant hazard (no change over time)
- **Weibull/Gompertz:** Monotonically increasing or decreasing hazard
- **Log-logistic/Log-normal:** Non-monotonic hazard (can increase then decrease)

- **Gamma/Generalized Gamma:** Flexible hazard shapes

The range of NMB across distributions indicates sensitivity to these extrapolation assumptions. A large range suggests that long-term survival projections are particularly influential and that the choice of parametric form matters for decision-making.

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